Workshop 4

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Evidencia

```
casti@MLC MINGW64 ~/workshop3-docker
$ cd ~

casti@MLC MINGW64 ~
$ mkdir -p workshop4-docker/postgres/init

casti@MLC MINGW64 ~
$ cd workshop4-docker
```

Parte 1: Construcción manual

```
casti@MLC MINGW64 ~/workshop4-docker
$ touch docker-compose.yml

casti@MLC MINGW64 ~/workshop4-docker
$ notepad docker-compose.yml
```

```
services:
  postgres:
    image: postgres:16
    environment:
      POSTGRES_DB: universidad
      POSTGRES_USER: admin
      POSTGRES_PASSWORD: admin
    ports:
      - "5432:5432"
```

```
casti@MLC MINGW64 ~/workshop4-docker
$ docker compose up -d
[+] up 2/2
 ✔ Network workshop4-docker_default      Created
 ✔ Container workshop4-docker-postgres-1 Started

casti@MLC MINGW64 ~/workshop4-docker
$ docker compose ps
NAME                          IMAGE         COMMAND                  SERVICE    CREATED          STATUS          PORTS
workshop4-docker-postgres-1   postgres:16   "docker-entrypoint.s…"   postgres   12 seconds ago   Up 12 seconds   0.0.0.0:5432->5432/tcp, [::]:5432->5432/tcp
```

Entrar a PSQL

```
casti@MLC MINGW64 ~/workshop4-docker
$ winpty docker compose exec postgres psql -U admin -d universidad
psql (16.15 (Debian 16.15-1.pgdg13+2))
Type "help" for help.
```

Parte 2: Crear manualmente la estructura

```
casti@MLC MINGW64 ~/workshop4-docker
$ winpty docker compose exec postgres psql -U admin -d universidad
psql (16.15 (Debian 16.15-1.pgdg13+2))
Type "help" for help.

universidad=# CREATE SCHEMA academia;
CREATE SCHEMA
universidad=# CREATE TABLE academia.students (
universidad(#     id SERIAL PRIMARY KEY,
universidad(#     name VARCHAR(100) NOT NULL,
universidad(#     email VARCHAR(120) UNIQUE NOT NULL
universidad(# );
CREATE TABLE
universidad=# CREATE TABLE academia.courses (
universidad(#     id SERIAL PRIMARY KEY,
universidad(#     name VARCHAR(100) NOT NULL,
universidad(#     credits INTEGER NOT NULL
universidad(# );
CREATE TABLE
universidad=# INSERT INTO academia.students (name, email) VALUES
universidad-# ('Ana', 'ana@example.com'),
universidad-# ('Luis', 'luis@example.com'),
universidad-# ('Lucia', 'lucia@example.com');
INSERT 0 3
universidad=# INSERT INTO academia.courses (name, credits) VALUES
universidad-# ('Cloud Computing', 3),
universidad-# ('Internet of Things', 4);
INSERT 0 2
universidad=# SELECT * FROM academia.students;
 id | name  |       email
----+-------+-------------------
  1 | Ana   | ana@example.com
  2 | Luis  | luis@example.com
  3 | Lucia | lucia@example.com
(3 rows)

universidad=# SELECT * FROM academia.courses;
 id |        name        | credits
----+--------------------+---------
  1 | Cloud Computing    |       3
  2 | Internet of Things |       4
(2 rows)
```

```
universidad=# \dn
       List of schemas
   Name   |       Owner
----------+-------------------
 academia | admin
 public   | pg_database_owner
(2 rows)

universidad=# \dt academia.*
          List of relations
  Schema  |   Name   | Type  | Owner
----------+----------+-------+-------
 academia | courses  | table | admin
 academia | students | table | admin
(2 rows)

universidad=# \q
```

Parte 3: Automatización

```
casti@MLC MINGW64 ~/workshop4-docker
$ touch postgres/init/01-schema.sql

casti@MLC MINGW64 ~/workshop4-docker
$ notepad postgres/init/01-schema.sql
```

```
CREATE SCHEMA IF NOT EXISTS academia;

CREATE TABLE IF NOT EXISTS academia.students (
    id SERIAL PRIMARY KEY,
    name VARCHAR(100) NOT NULL,
    email VARCHAR(120) UNIQUE NOT NULL
);

CREATE TABLE IF NOT EXISTS academia.courses (
    id SERIAL PRIMARY KEY,
    name VARCHAR(100) NOT NULL,
    credits INTEGER NOT NULL
);
```

```
casti@MLC MINGW64 ~/workshop4-docker
$ touch postgres/init/02-data.sql

casti@MLC MINGW64 ~/workshop4-docker
$ notepad postgres/init/02-data.sql
```

```
INSERT INTO academia.students (name, email) VALUES
('Ana', 'ana@example.com'),
('Luis', 'luis@example.com'),
('Marta', 'marta@example.com');

INSERT INTO academia.courses (name, credits) VALUES
('Cloud Computing', 3),
('Internet of Things', 4);
```

ACLARACIÓN: Acá tuve un errorcito y me tocó modificar a Lucía porque realmente era Marta…

```
casti@MLC MINGW64 ~/workshop4-docker
$ docker compose exec postgres psql -U admin -d universidad -c "UPDATE academia.students SET name = 'Marta', email = 'marta@example.com' WHERE id = 3;"
UPDATE 1

casti@MLC MINGW64 ~/workshop4-docker
$ docker compose exec postgres psql -U admin -d universidad -c "SELECT * FROM academia.students;"
 id | name  |       email
----+-------+-------------------
  1 | Ana   | ana@example.com
  2 | Luis  | luis@example.com
  3 | Marta | marta@example.com
(3 rows)
```

Parte 4: Montar los scripts

```
casti@MLC MINGW64 ~/workshop4-docker
$ notepad docker-compose.yml
```

```yaml
services:
  postgres:
    image: postgres:16
    environment:
      POSTGRES_DB: universidad
      POSTGRES_USER: admin
      POSTGRES_PASSWORD: admin
    ports:
      - "5432:5432"
    volumes:
      - postgres_data:/var/lib/postgresql/data
      - ./postgres/init:/docker-entrypoint-initdb.d:ro

volumes:
  postgres_data:
```

Parte 5: Regla importante de inicialización

```
casti@MLC MINGW64 ~/workshop4-docker
$ docker compose down -v
[+] down 2/2
 ✔ Container workshop4-docker-postgres-1 Removed
 ✔ Network workshop4-docker_default      Removed

casti@MLC MINGW64 ~/workshop4-docker
$ docker compose up -d
[+] up 3/3
 ✔ Network workshop4-docker_default      Created
 ✔ Volume workshop4-docker_postgres_data Created
 ✔ Container workshop4-docker-postgres-1 Started

casti@MLC MINGW64 ~/workshop4-docker
$ docker compose ps
NAME                          IMAGE          COMMAND                  SERVICE    CREATED         STATUS         PORTS
workshop4-docker-postgres-1   postgres:16    "docker-entrypoint.s…"   postgres   5 seconds ago   Up 5 seconds   0.0.0.0:5432->5432/tcp, [::]:5432->5432/tcp
```

Verificación de que todo se creó solo:

```
casti@MLC MINGW64 ~/workshop4-docker
$ docker compose exec postgres psql -U admin -d universidad -c "\dn"
      List of schemas
   Name    |       Owner
-----------+-------------------
 academia  | admin
 public    | pg_database_owner
(2 rows)


casti@MLC MINGW64 ~/workshop4-docker
$ docker compose exec postgres psql -U admin -d universidad -c "\dt academia.*"
          List of relations
  Schema  |   Name   | Type  | Owner
----------+----------+-------+-------
 academia | courses  | table | admin
 academia | students | table | admin
(2 rows)

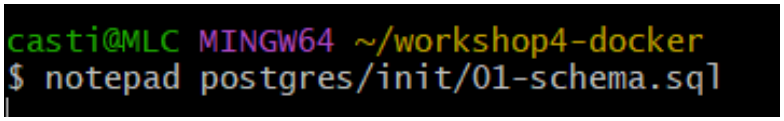

casti@MLC MINGW64 ~/workshop4-docker
$ docker compose exec postgres psql -U admin -d universidad -c "SELECT * FROM academia.students;"
 id | name  |       email
----+-------+-------------------
  1 | Ana   | ana@example.com
  2 | Luis  | luis@example.com
  3 | Marta | marta@example.com
(3 rows)


casti@MLC MINGW64 ~/workshop4-docker
$ docker compose exec postgres psql -U admin -d universidad -c "SELECT * FROM academia.courses;"
 id |        name        | credits
----+--------------------+---------
  1 | Cloud Computing    |       3
  2 | Internet of Things |       4
(2 rows)
```

```
casti@MLC MINGW64 ~/workshop4-docker
$ docker compose restart
[+] restart 0/1
 - Container workshop4-docker-postgres-1 Restarting

casti@MLC MINGW64 ~/workshop4-docker
$ docker compose exec postgres psql -U admin -d universidad -c "SELECT * FROM academia.students;"
 id | name  |       email
----+-------+-------------------
  1 | Ana   | ana@example.com
  2 | Luis  | luis@example.com
  3 | Marta | marta@example.com
(3 rows)
```

Reto – Parte 6



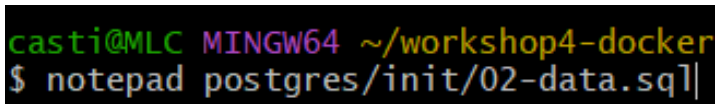

```
CREATE SCHEMA IF NOT EXISTS academia;

CREATE TABLE IF NOT EXISTS academia.students (
    id SERIAL PRIMARY KEY,
    name VARCHAR(100) NOT NULL,
    email VARCHAR(120) UNIQUE NOT NULL
);

CREATE TABLE IF NOT EXISTS academia.teachers (
    id SERIAL PRIMARY KEY,
    name VARCHAR(100) NOT NULL,
    email VARCHAR(120) UNIQUE NOT NULL
);

CREATE TABLE IF NOT EXISTS academia.courses (
    id SERIAL PRIMARY KEY,
    name VARCHAR(100) NOT NULL,
    credits INTEGER NOT NULL,
    teacher_id INTEGER REFERENCES academia.teachers(id)
);

CREATE TABLE IF NOT EXISTS academia.enrollments (
    id SERIAL PRIMARY KEY,
    student_id INTEGER NOT NULL REFERENCES academia.students(id),
    course_id INTEGER NOT NULL REFERENCES academia.courses(id),
    enrolled_at DATE NOT NULL DEFAULT CURRENT_DATE,
    UNIQUE (student_id, course_id)
);
```



```sql
INSERT INTO academia.students (name, email) VALUES
('Ana', 'ana@example.com'),
('Luis', 'luis@example.com'),
('Marta', 'marta@example.com');

INSERT INTO academia.teachers (name, email) VALUES
('Carlos Ramirez', 'carlos@example.com'),
('Diana Torres', 'diana@example.com');

INSERT INTO academia.courses (name, credits, teacher_id) VALUES
('Cloud Computing', 3, 1),
('Internet of Things', 4, 1),
('Bases de Datos', 3, 2);

INSERT INTO academia.enrollments (student_id, course_id) VALUES
(1, 1),
(1, 2),
(2, 1),
(3, 2),
(3, 3);
```

Reconstruir

```
casti@MLC MINGW64 ~/workshop4-docker
$ docker compose down -v
[+] down 3/3
 ✔ Container workshop4-docker-postgres-1 Removed
 ✔ Volume workshop4-docker_postgres_data Removed
 ✔ Network workshop4-docker_default      Removed

casti@MLC MINGW64 ~/workshop4-docker
$ docker compose up -d
[+] up 3/3
 ✔ Network workshop4-docker_default      Created
 ✔ Volume workshop4-docker_postgres_data Created
 ✔ Container workshop4-docker-postgres-1 Started

casti@MLC MINGW64 ~/workshop4-docker
$ docker compose ps
NAME                        IMAGE          COMMAND                  SERVICE    CREATED          STATUS         PORTS
workshop4-docker-postgres-1   postgres:16   "docker-entrypoint.s…"   postgres   10 seconds ago   Up 9 seconds   0.0.0.0:5432->5432/tcp, [::]:5432->5432/tcp
```

Verificar

```
casti@MLC MINGW64 ~/workshop4-docker
$ docker compose exec postgres psql -U admin -d universidad -c "\dt academia.*"
        List of relations
  Schema  |    Name     | Type  | Owner
----------+-------------+-------+-------
 academia | courses     | table | admin
 academia | enrollments | table | admin
 academia | students    | table | admin
 academia | teachers    | table | admin
(4 rows)


casti@MLC MINGW64 ~/workshop4-docker
$ docker compose exec postgres psql -U admin -d universidad -c "SELECT * FROM academia.teachers;"
 id |      name      |        email
----+----------------+---------------------
  1 | Carlos Ramirez | carlos@example.com
  2 | Diana Torres   | diana@example.com
(2 rows)


casti@MLC MINGW64 ~/workshop4-docker
$ docker compose exec postgres psql -U admin -d universidad -c "SELECT * FROM academia.courses;"
 id |        name        | credits | teacher_id
----+--------------------+---------+------------
  1 | Cloud Computing    |       3 |          1
  2 | Internet of Things |       4 |          1
  3 | Bases de Datos     |       3 |          2
(3 rows)


casti@MLC MINGW64 ~/workshop4-docker
$ docker compose exec postgres psql -U admin -d universidad -c "SELECT * FROM academia.enrollments;"
 id | student_id | course_id | enrolled_at
----+------------+-----------+-------------
  1 |          1 |         1 | 2026-09-06
  2 |          1 |         2 | 2026-09-06
  3 |          2 |         1 | 2026-09-06
  4 |          3 |         2 | 2026-09-06
  5 |          3 |         3 | 2026-09-06
(5 rows)


casti@MLC MINGW64 ~/workshop4-docker
$ docker compose exec postgres psql -U admin -d universidad -c "SELECT s.name AS estudiante, c.name AS curso, t.name AS profesor FROM academia.enrollments e JOIN academia.students s ON s.id = e.student_id JOIN
academia.courses c ON c.id = e.course_id JOIN academia.teachers t ON t.id = c.teacher_id ORDER BY s.name;"
 estudiante |        curso        |    profesor
------------+---------------------+----------------
 Ana        | Cloud Computing     | Carlos Ramirez
 Ana        | Internet of Things  | Carlos Ramirez
 Luis       | Cloud Computing     | Carlos Ramirez
 Marta      | Internet of Things  | Carlos Ramirez
 Marta      | Bases de Datos      | Diana Torres
(5 rows)
```

```
casti@MLC MINGW64 ~/workshop4-docker
$ docker compose exec postgres psql -U admin -d universidad -c "INSERT INTO academia.enrollments (student_id, course_id) VALUES (99, 1);"
ERROR:  insert or update on table "enrollments" violates foreign key constraint "enrollments_student_id_fkey"
DETAIL:  Key (student_id)=(99) is not present in table "students".

What's next:
    Debug this Compose error with Gordon → docker ai "help me fix this compose error"
```

```
casti@MLC MINGW64 ~/workshop4-docker
$ docker compose down
[+] down 2/2
 ✔ Container workshop4-docker-postgres-1 Removed
 ✔ Network workshop4-docker_default      Removed
```